

ECEN 662: Estimation & Detection Theory

Lecture 17: Transform and Rejection Sampling

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Sampling/random variable generation/simulation

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- Here we shall focus on two of the most basic sampling methods: *transformation* and *rejection*
- Our main criterion for evaluating different sampling algorithms are the *required knowledge* of the distribution and *algorithm efficiency*

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- Thus, in order to sample from a CDF F on $\mathbb{R}$, it suffices to generate a random variable $U \sim \mathcal{U}([0, 1])$ and make the transformation $x = F^{-1}(u)$
- For example, if $X \sim \text{Exp}(1)$, then $F(x) = 1 - e^{-x}$ and $F^{-1}(u) = -\log(1 - u)$. Therefore, if $U \sim \mathcal{U}([0, 1])$, then $X = -\log(U) \sim \text{Exp}(1)$ (as U and $1 - U$ are both uniform between 0 and 1)

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- It follows immediately that

$$\mathbb{P}(F^{-1}(U) \leq x) = \mathbb{P}(U \leq F(x)) = F(x)$$

where the last equality follows from the fact that $U \sim \mathcal{U}([0, 1])$ and $F(x) \in [0, 1]$

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- Most noticeably, the CDF of the standard Gaussian distribution *cannot* be expressed explicitly, since

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- Thus, to use the inverse transform to generate a standard Gaussian random variable, one may have to use an *approximate* expression for the generalized inverse Φ^{-1}

- One known such approximation is

$$\Phi^{-1}(u) \approx t - \frac{2.30753 + 0.27061t}{1 + 0.99299t + 0.04481t^2}$$

where

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- If no other fast sampling method was available, this approximation could be used in settings which do *not* require much precision in the *tails* of $\mathcal{N}(0, 1)$

General transformation method

- While the inverse transform offers a *generic* link between an arbitrary random variable on $\mathbb{R}$ and the uniform random variable, sometimes there is a *special* link between a specific random variable to some random variables that are easy to simulate. This special link can often be exploited to sample from the targeted distribution

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- For example, let X_1 and X_2 be two independent standard Gaussian random variables. It is known that the polar coordinates (R, Θ) of (X_1, X_2) are distributed as:

$$R \sim \text{Exp}(1/2) \quad \text{and} \quad \Theta \sim \mathcal{U}([0, 2\pi])$$

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- Therefore, if U_1 and U_2 are i.i.d. $\mathcal{U}([0, 1])$

$$X_1 = \sqrt{-2 \log U_1} \cos(2\pi U_2), \quad X_2 = \sqrt{-2 \log U_1} \sin(2\pi U_2)$$

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- Alternatively, if U and V are i.i.d. $\mathcal{U}([0, 1])$, the distribution of

$$\frac{U^{1/\alpha}}{U^{1/\alpha} + V^{1/\beta}}$$

conditioned on

$$U^{1/\alpha} + V^{1/\beta} \leq 1$$

is $Beta(\alpha, \beta)$

- However, given the constraint on $U^{1/\alpha} + V^{1/\beta}$, this result does *not* provide a good algorithm to generate $Beta(\alpha, \beta)$ random variables for *large* values of α and β

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- Here we present a very *clever* method known as the *acceptance-rejection* method
- The basic idea is to find an alternative probability distribution G , with density function g , from which we already have an efficient algorithm for sampling from (e.g., inverse transform method or whatever), but also such that g is “close” to the target density function f

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- The basic idea is to find an alternative probability distribution G , with density function g , from which we already have an efficient algorithm for sampling from (e.g., inverse transform method or whatever), but also such that g is “close” to the target density function f
- In particular, we assume that the ratio $f(x)/g(x)$ is *bounded* by a constant $M > 0$:

$$\sup_{x \in \mathcal{X}} \frac{f(x)}{g(x)} \leq M$$

- Here is the algorithm for generating a random variable X distributed as F :
 - 1) Generate a random variable Y according to the distribution G
 - 2) Generate a random variable U that is uniform between 0 and 1 and independent of Y
 - 3) If

$$U \leq \frac{f(Y)}{Mg(Y)}$$

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- Next we provide a few remarks in regard to the algorithm before proving that it actually works

Remarks

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- The success probability p can be calculated as follows:

$$\mathbb{P}\left(U \leq \frac{f(Y)}{Mg(Y)} \middle| Y = y\right) = \frac{f(y)}{Mg(y)}$$

and hence

$$p = \int_{\mathcal{X}} \frac{f(y)}{Mg(y)} g(y) dy = \frac{1}{M} \int_{\mathcal{X}} f(y) dy = \frac{1}{M}$$

- Thus, on average the number of iterations required is given by

$$p^{-1} = M$$

and we can now indeed see that it is desirable to choose our alternative density g so as to minimize this constant

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- In the context of the acceptance-rejection method, F is usually known as the *target* distribution, and G is known as the *proposal* distribution
- In the end, we obtain our X as having the *conditional* distribution of Y given that the event $U \leq \frac{f(Y)}{Mg(Y)}$ occurs

Proof that the algorithm works

- Let $B \subseteq \mathcal{X}$. Note that

$$\begin{aligned}\mathbb{P}\left(U \leq \frac{f(Y)}{Mg(Y)}, Y \in B\right) &= \int_B \mathbb{P}\left(U \leq \frac{f(y)}{Mg(y)}\right) g(y) dy \\ &= \int_B \frac{f(y)}{Mg(y)} g(y) dy \\ &= \frac{1}{M} \int_B f(y) dy\end{aligned}$$

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- It follows that

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- That is, the conditional distribution of Y given $U \leq \frac{f(Y)}{Mg(Y)}$ is indeed F

Example: Standard Gaussian distribution

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- To sample from the above distribution, the proposal distribution must have a *heavier* tail so that the density ratio $f(x)/g(x)$ is *bounded*
- One possible choice for the proposal distribution is the *two-sided exponential* distribution

$$g(x) = \frac{1}{2} \exp(-|x|)$$

which we already know how to sample from using the *inverse transform* method

- The density ratio

$$\frac{f(x)}{g(x)} = 2A \exp\left(|x| - \frac{x^2}{2}\right)$$

is *maximized* at $x = \pm 1$, giving

$$M = \sup_x \frac{f(x)}{g(x)} = 2 \exp(1/2)A$$

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- The *acceptance* rule is thus given by

$$U \leq \frac{f(Y)}{Mg(Y)} = \exp \left(|Y| - \frac{Y^2 + 1}{2} \right) = \exp \left(-\frac{(|Y| - 1)^2}{2} \right)$$

or equivalently

$$-\log U \geq \frac{(|Y| - 1)^2}{2}$$

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- Note that $-\log(U) \sim \text{Exp}(1)$, so in each iteration of the algorithm, we can simply use two independent *exponential* random variables to perform simulation

- Last but not the least, we see from this example that rejection sampling *only* requires the density function of the target distribution up to a *multiplicative* constant: The same constant *cancels out* in $f(Y)$ and M !

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- This makes makes rejection sampling potentially *suitable* for sampling from *posterior* distributions in Bayesian statistics

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- However, as the dimensions of the problem get larger, the ratio of the embedded volume to the “corners” of the embedding volume tends towards zero, thus a lot of rejections can take place before a useful sample is generated, thus making the algorithm inefficient and impractical
- In high dimensions, it is necessary to use a different approach, typically a *Markov chain Monte Carlo* method such as Metropolis sampling or Gibbs sampling